An Artifact for the Article:

“On the Vulnerability Proneness of Multilingual Code”

ABSTRACT

This artifact contains functional versions of the toolkit PolyFax and corresponding datasets for the empirical study. We have published and maintained the code of PolyFax on GitHub and as a Docker image on Docker Hub with the datasets included and all of the dependencies for PolyFax installed. Additionally, users can also choose to build PolyFax from source code and retrieve the dataset from Figshare. A host machine’s suggested requirements are at least 32 GB of hard disk space and 16 GB of memory for reproducing the results in the paper.

Our empirical experiments presented in the paper can be divided into three parts. The first part is the results of the two tools (i.e., VCC and LIC as described in Sections 2.3 and 2.4 in the paper) developed for the study. The results of the tools are contained in the dataset as the intermediate results for the paper. The second part is the NBR analysis results on security vulnerability between language selection, as presented in the results of RQ1, which includes three tables (i.e., Tables 3,4,5). The third part is the NBR analysis results on security vulnerability between language interfacing mechanism and individual language, as presented in the results of RQ2, which includes three tables (i.e., Tables 6,7,8).

As the study refers to the analysis of over 10 K repositories and over 20 million commits, it can take over one week to finish all the steps of the experiments. Thus, to make it easier for users to reproduce the experiments, we have saved the intermediate results of the tools VCC and LIC, and the users can run the NBR analysis directly. Nevertheless, our tools are purposely made highly reusable; thus, we also published the toolkit PolyFax for users to run the complete experiments against other projects than those in our paper, starting from data grabbing all the way through the NBR analysis.

Besides, our study also involves manual reviews and user studies. The raw data and results are also provided in our artifact.

Further details regarding the content of this artifact and how to install and use it can be found in the README included in this package.